

Study of Bridge Design

ENGS121 Mechanics Lab Section D

13 May, 2025

1. Introduction

The experiment is done to study the design of cable-stay bridges, focusing on the behavior of tension forces in the suspension system of the bridge. Experiment is divided into parts. Examining how the tension in cable changes after altering the hanging mass. Examining the influence of the position of the cable attachment on the tension force. Lastly, studying the relationship between two pairs of tension forces in the cables attached at two different points. Since improper balance or resonance can cause serious structural failures, the main point of the experiment is to better understand the behavior of cable stayed bridges through theoretical analysis and experimental measurements.

2. Measurements

Instruments

Name	Quantity	Properties
Stand	1 pc	Stand

Dynamometer(s)	1 pc	Resolution of 0.1 N
Weights	Set of Lab Weights	Different weights
Scales	1 pc	With an accuracy of 0.001 kg
Ruler	1pc	Resolution of 0.01 m
Bridge Deck	1 pc	Mass: 0.346 kg Length: 0.600 m

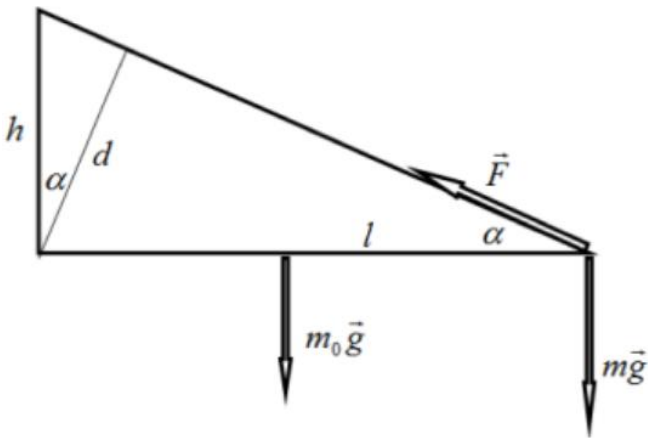
Errors

Source of error	Type	Evaluation
Friction in the cables or supports	Random	Friction between the cables and their supports can alter the measured tension force.
Misalignment of the Bridge Model	Systematic	Some parts of the equipment used might be not so perfectly aligned with their designated properties which can cause errors in measurements.
Forces in different axis	Systematic	While doing the experiment the forces might not align in the same plane, especially in the top of the bridge where the dynamometer is attached, this can alter the results of the observation.

Variables

variable	Value	Resolution
Attached weight	Controlled	1 g
Places of attached weights and dynamometer	Controlled	1 mm
Values of dynamometer	Controlled/Measured	0.1 N

Expressing the Theoretical Formula



In order to keep the bridge in a fixed position aka equilibrium the net of the forces that influence on the bridge must be 0. After considering the left edge of the bridge as a pivot, and counterclockwise direction as positive.

$$\sum \tau = 0$$

$$\tau_t - \tau_0 - \tau = 0$$

$$Fd - m_0 g l_{CM} - m g l = 0$$

$$F = \frac{m_0 g l_{CM} + m g l}{d} = \frac{g l}{d} m + \frac{m_0 g l_{CM}}{d}. \text{ (Linearized)}$$

Part A

In Part A the goal is to determine the dependence of tension F on m experimentally. The lever of force F can be calculated by simply just plugging m = 0 into the above derived formula, hence

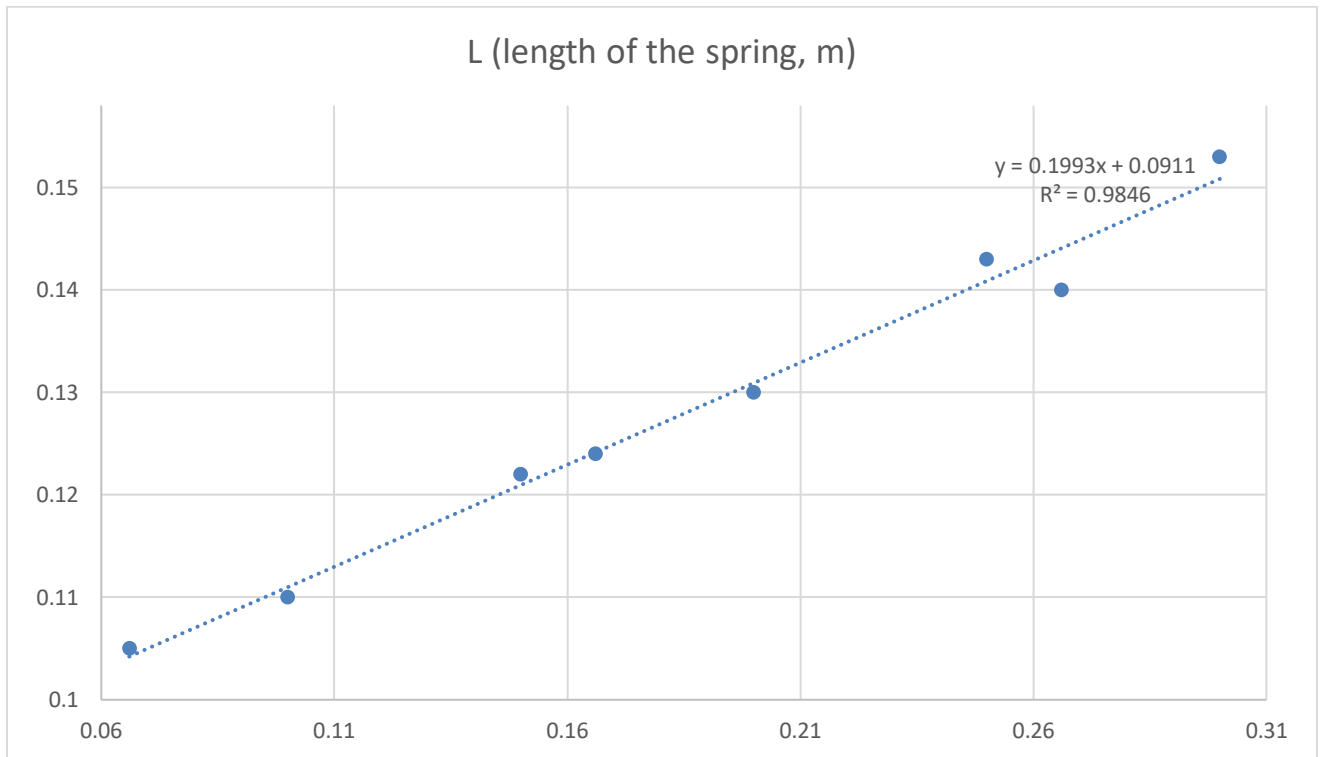
$d = \frac{m_0 g l_{CM}}{F}$, F when m = 0 is measured to be 3.8 N, the center of mass was 0.31 m right from the

pivot. After substituting and calculating **d ≈ 0.275 m** (lever of force F).

L (length of spring, m)	m (hanged mass, kg)	l (distance to mass hanging point, m)	x (cm)	d (cm)
0.105	0.066	0.4	0.27	0.275
0.110	0.1	0.4		0.275
0.122	0.15	0.4	0.27	0.275
0.124	0.166	0.4	0.27	0.275
0.13	0.2	0.4	0.27	0.275

0.143	0.25	0.4	0.27	0.275
0.14	0.266	0.4	0.27	0.275
0.153	0.3	0.4	0.27	0.275

Dependence of L on m



$$F = \frac{m_0 g l_{CM} + m g l}{d} = \frac{g l}{d} m + \frac{m_0 g l_{CM}}{d}. \quad F = k L \mid (F = k x). \Rightarrow L = \frac{g l}{k d} m + \frac{m_0 g l_{CM}}{k d}$$

The formula matches the form of a linear equation $L = \text{slope} * m + \text{intercept}$, where $\text{slope} = \frac{g l}{k d}$

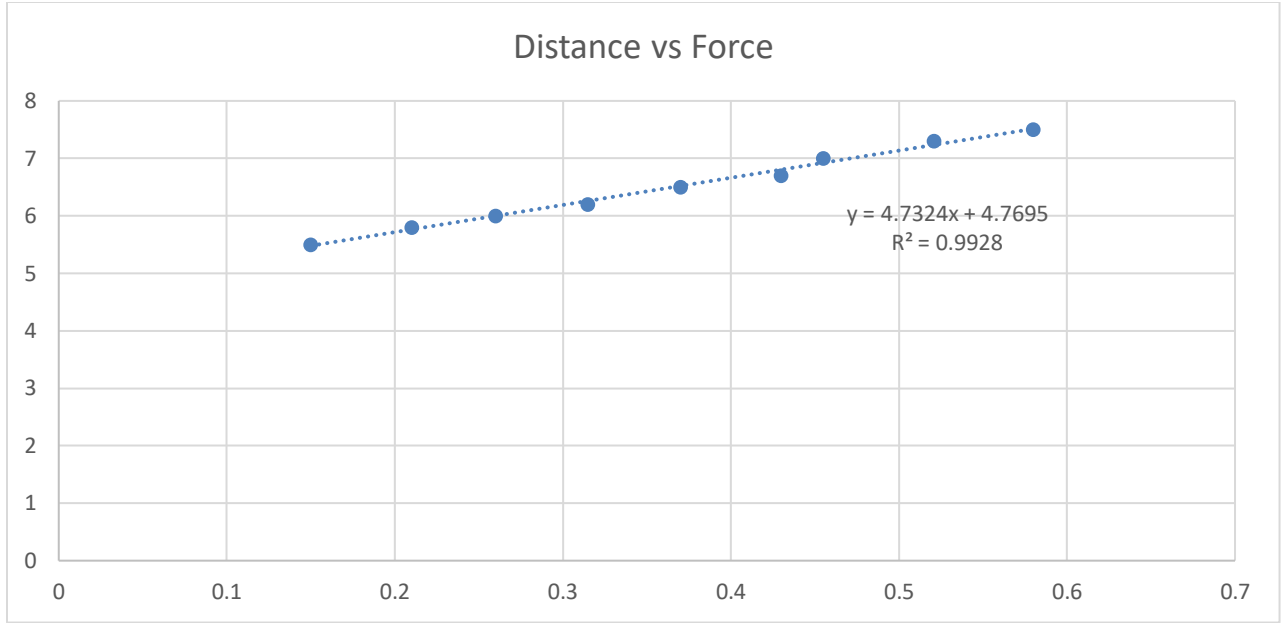
and intercept is $\frac{m_0 g l_{CM}}{k d}$. Correspondingly $L = 0.1993m + 0.0911$.

$$k = \frac{g l}{a d}, \text{ (where } a = \text{slope)} \Rightarrow k = (9.81 * 0.4) / (0.1993 * 0.275) = 71.6 \text{ N (stiffness of the spring)}$$

Part B

The aim of part B is to determine the dependence of F on l experimentally. Recall the theoretical formula can be given in the following form: $F = \frac{m_0 g l_{CM} + m g l}{d} = \frac{g m}{d} l + \frac{m_0 g l_{CM}}{d}$, where the constant mass $m = 0.087 \text{ kg}$ is hung from different positions. The position of the dynamometer did not change, so $d = \mathbf{0.183 \text{ m}}$.

F (reading of dynamometer) N	m (hanged mass) kg	X, m
5.5	0.087	0.15
5.8	0.087	0.21
6	0.087	0.26
6.2	0.087	0.315
6.5	0.087	0.37
6.7	0.087	0.43
7	0.087	0.455
7.3	0.087	0.521
7.5	0.087	0.58



As was assumed, the graph again appears to be linear. Let us compare the slope and intercept with their theoretical values.

First, let us find the error in the slope:

$$(\Delta \text{slope})^2 = \frac{1}{n(\overline{x^2} - \bar{x}^2)} \frac{\sum (y_{\text{predicted}} - y_{\text{experiment}})^2}{n-2}.$$

Where $\Delta \text{slope} \approx 0.1 \text{ N/m}$. The theoretical value of the slope is $\frac{gm}{d} = \frac{9.80 \times 0.087}{0.183} \approx 4.66 \text{ N/m}$,

which lies in the $[\text{slope} - \Delta \text{slope}, \text{slope} + \Delta \text{slope}] = [4.62, 4.82]$ interval.

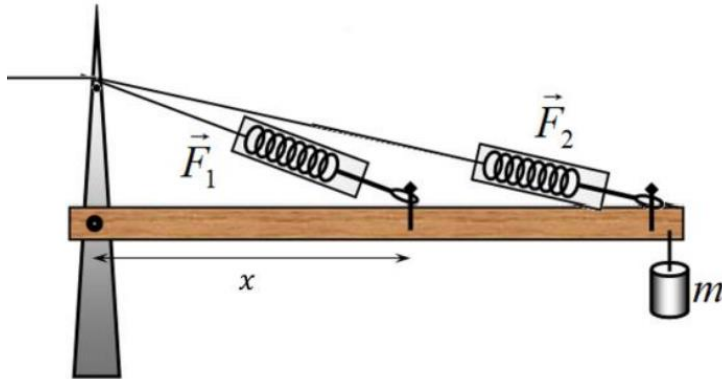
$\Delta(\text{intercept})^2 = \overline{x^2} \times (\Delta k)^2$. $\Delta(\text{intercept}) \approx 0.040 \text{ N}$. The theoretical value of the intercept

is $\frac{m_0 g l_{CM}}{d} = \frac{0.29 \times 9.80 \times 0.309}{0.181} \approx 4.85 \text{ N}$, which lies in the interval $[\text{intercept} -$

$2\Delta \text{intercept}, \text{intercept} + 2\Delta \text{intercept}] = [4.693, 4.853]$. **Based on these calculations, it can**

be concluded that the theoretical formula is consistent.

Part C



In part C, two dynamometers are attached to the deck. The aim is to determine the dependence between F_1 and F_2 and to find the optimal forces ensuring maximum safety for the bridge.

First, let us try to derive the theoretical formula. The bridge is in equilibrium: hence the net torque is 0. Take counterclockwise direction as positive the left edge of the bridge as pivot point.

$$\sum \tau = 0$$

$$\tau_1 + \tau_2 - \tau_0 - \tau = 0$$

$$F_1 d_1 + F_2 d_2 = m_0 g l_{CM} + m g l.$$

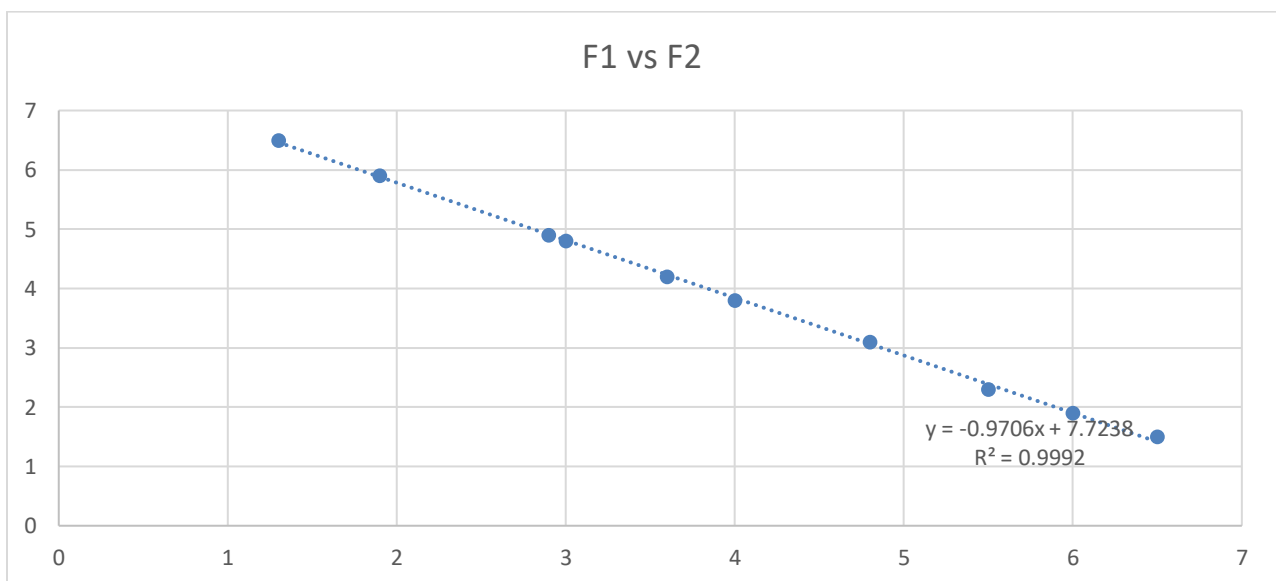
The argument is F_1 . $F_2 = -\frac{d_1}{d_2} F_1 + \frac{m_0 g l_{CM} + m g l}{d_2}$.

$m = 0.100 \text{ kg}$, $l = 0.528 \text{ m}$, $h = 0.20 \text{ m}$. Using $x_1 = 0.466 \text{ m}$, $x_2 = 0.582 \text{ m}$ and geometry,

$$d_1 \approx 0.176 \text{ m}, d_2 \approx 0.181 \text{ m}.$$

F1 (control) N	F2 (measure) N
1.3	6.5

1.9	5.9
2.9	4.9
3	4.8
3.6	4.2
4	3.8
4.8	3.1
5.5	2.3
6	1.9
6.5	1.5



$$(\Delta slope)^2 = \frac{1}{n(\overline{x^2} - \bar{x}^2)} \frac{\sum (y_{predicted} - y_{experiment})^2}{n-2}.$$

$\Delta slope \approx 0.0097$. The theoretical value of the slope is $-\frac{d_1}{d_2} = -\frac{0.176}{0.181} \approx -0.972$, which lies in the $[slope - \Delta slope, slope + \Delta slope] = [-0.980, -0.961]$ interval.

Now, let us find the uncertainty in the intercept: $\Delta(intercept)^2 = \overline{x^2} \times (\Delta k)^2$.

$\Delta(intercept) \approx 0.042 \text{ N}$. The theoretical value of the intercept is $\frac{m_0 g l_{CM} + m g l}{d_2} \approx 7.71 \text{ N}$, which lies in the interval $[intercept - \Delta intercept, intercept + \Delta intercept] = [7.68, 7.77]$. **Based on these calculations, it can be concluded that the theoretical formula is consistent.**